

**Report
Of
Python-Based Application Development
and Automation**

Submitted
To

**School of Computer Science and Engineering
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of
B.Tech CSE



By

Roll Number of Student: 25SCS1003004635

Name of Student: SUDHANSHU KUMAR

Batch: 2026-27



CANDIDATE'S DECLARATION

I, SUDHANSHU KUMAR, Roll No. 25SCS1003004635, do hereby declare that the internship report titled “Python-Based Application Development and Automation” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering, The internship was undertaken as a Python Developer Intern at Codec Technologies Pvt. Ltd. from 10/06/2026 to 10/07/2026. All references used in preparing this report have been acknowledged. I affirm that this report represents my academic work and has not been submitted to any other institute for any degree/diploma requirement, and I shall be responsible for any copyright violation in this regard.

Signature

Student Name: SUDHANSHU KUMAR

Roll No.: 25SCS1003004635

ACKNOWLEDGEMENT

I take this opportunity to express my sincere gratitude to everyone who supported me throughout my internship program at Codec Technologies Pvt. Ltd. I am thankful for the guidance, learning resources, constructive feedback, and encouragement provided during the internship. The internship gave me practical exposure to Python development and helped me understand how programming concepts can be applied to Intern at Codec Technologies Pvt. Ltd. from 10/06/2026 to 10/07/2026. All references used in preparing this report have been acknowledged.

I would also like to thank the School of Computer Science and Engineering, IILM University, Greater Noida, for providing me with the opportunity and academic support to undertake this internship as part of my B.Tech CSE curriculum.

Finally, I express my heartfelt gratitude to my parents, teachers, and friends for their continuous support, encouragement, and blessings throughout this endeavour.

Signature

Student Name: SUDHANSHU KUMAR

Roll No.: 25SCS1003004635

INTERNSHIP OFFER LETTER

Dear Sudhanshu kumar,

We are pleased to offer you a 1 Month Internship as Project Intern at Codec Technologies, a global platform dedicated to empowering learners and connecting diverse talent to create meaningful career opportunities worldwide. Codec Technologies delivers dedicated IT and business consultancy services across 27+ countries, empowering global innovation and strategic growth.

Internship Details:

- Designation : Python Developer Intern
- Location & Mode: Pan India, Hybrid
- Duration : 10/06/2026 to 10/07/2026
- Reporting to : Assigned Project Head(s)

This program provides an immersive experience, enabling you to gain industry-level knowledge and acquire valuable skills. Exceptional interns may also qualify for a scholarship award based on weekly performance reviews conducted by project supervisor.

If you accept this offer and the terms above, please complete your assigned projects and training tasks on our platform and via email.

Best regards,

Sincerely,

Dr. Anurag Shrivastava,
Codec Technologies India
NCS ID: E19E86-0116588288923
Date Of Issue - 10/06/2026



Dr. Anurag Shrivastava
Talent Acquisition Manager



Google for Education
Partner



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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

Python is a widely used programming language for developing applications, automating repetitive tasks, processing data, and building intelligent software solutions. During the two-month internship at Codec Technologies Pvt. Ltd., the focus was on Python development and practical application of programming concepts. The internship period was from 20/06/2026 to 20/08/2026, as stated in the supplied internship certificate.

The project-oriented work covered the design of a Python-based application workflow in which input data is collected, processed through Python logic, validated, and converted into useful outputs. The work provided practical exposure to programming structures, functions, modules, data handling, error management, testing, and basic automation. The experience complemented the technical curriculum of a Computer Science and Engineering programme.

4.2 Organization Profile

Codec Technologies describes itself as an IT and business consultancy and technology-learning organisation. Its official website states that its mission is to empower individuals and organisations through technology solutions and learning experiences, and that it provides internship and training programmes including Python Developer internships. The organisation also states that its programmes are designed to provide hands-on exposure to industry tools and workflows.

4.3 Problem Statement

Many routine software tasks involve repetitive data entry, validation, transformation, file handling, and generation of structured outputs. When these tasks are performed manually, they can consume time and may introduce avoidable errors. A Python-based solution can reduce repetitive work by applying consistent program logic to input data and producing processed results in a structured form.

The internship project therefore focused on understanding how Python can be used to create a practical application workflow that accepts input, performs validation and processing, handles exceptions, and generates useful output. The objective was to combine programming fundamentals with a small-scale application development approach suitable for learning and practical implementation.

4.4 Project Objectives

- To strengthen practical Python programming skills through application-oriented work.
- To design a simple Python workflow for input collection, processing, validation, and output generation.
- To apply functions, modules, conditional statements, loops, and exception handling in a structured application.
- To understand basic data handling and file-processing concepts using Python.
- To test the application using sample inputs and identify and correct logical errors.
- To understand professional development practices such as documentation, version control, and organised project structure.

4.5 Scope of the Project

The scope of the internship project covered Python programming, application logic, data processing, basic automation, testing, and documentation. The work was intended as a learning and prototype-level implementation rather than a large-scale production deployment. The project can be extended in the future with a graphical interface, database integration, web APIs, cloud deployment, and more advanced automation features.

4.6 Technologies and Tools Used

- Programming Language: Python for application logic and automation.
- Development Environment: Python-compatible IDE/editor and command-line tools.
- Data Handling: Python data structures and file-processing techniques.
- Libraries/Modules: Standard Python modules and relevant packages where required for implementation.
- Version Control: Git/GitHub concepts for source-code management and project tracking.
- Documentation & Presentation: Word-processing and presentation tools for preparing internship material.

4.7 System Architecture

The proposed Python application follows a simple input-processing-output architecture. A user or source provides input data to the application. The Python processing layer validates the input, applies the required business logic or transformation, and handles exceptions. The processed information is then stored or presented as an output. The modular structure allows individual components to be tested and improved independently.

INPUT	→	PYTHON PROCESSING	→	VALIDATION / ERROR HANDLING	→	OUTPUT
Data / User Input	→	Functions & Modules	→	Checks & Exceptions	→	Processed Result

Figure 4.1: System Architecture of the Python-Based Application

4.8 Methodology

- Orientation: Understanding the internship programme, Python development workflow, and project requirements.
- Requirement Analysis: Identifying the inputs, processing steps, validation rules, and expected outputs.
- Design: Breaking the application into logical functions/modules and planning the data flow.
- Development: Implementing the application using Python and structured programming practices.

4.8 Methodology (Continued)

- Data Processing: Applying Python logic to transform and validate sample data.
- Exception Handling: Testing invalid or unexpected inputs and handling errors without terminating the application unnecessarily.
- Testing: Running sample test cases, checking outputs, debugging issues, and refining the implementation.
- Documentation: Recording the project structure, working procedure, and learning outcomes.

4.9 Expected Outcomes

- A functional prototype demonstrating Python-based input processing and automation.
- Improved understanding of Python syntax, functions, modules, data handling, and exception handling.
- Practical experience in organising a small software project into manageable components.
- Better debugging and testing skills through repeated validation of sample inputs and outputs.
- Exposure to professional development practices including documentation and source-code management.
- Increased confidence in using Python for future academic and software-development projects.

CHAPTER 5: BIBLIOGRAPHY/REFERENCES

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3. AICTE Internship Portal, Codec Technologies - Python Developer Internship programme information. Accessed August 2026.
4. Internship Certificate, Codec Technologies Pvt. Ltd., awarded to Aryan Kumar, Python Developer Intern, conducted from 20/06/2026 to 20/08/2026.
5. Python Software Foundation, Python Documentation - reference material for Python programming concepts. Accessed August 2026.



CERTIFICATE OF INTERNSHIP

This certificate is awarded to
Sudhanshu kumar

From Codec Technologies Pvt. Ltd.

In recognition of his/her efforts and achievements in completing the 1 Month AICTE & ICAC

Approved internship program as

Python Developer Intern

Conducted from 10/06/2026 to 10/07/2026



AICTE ID - CORPORATE6759d549ce59e1733940553

A handwritten signature in black ink, appearing to read "Anurag", positioned above a horizontal line.

Dr. Anurag Shrivastava
Program Manager